ECONOMIC GROWTH AND SUSTAINABLE DEVELOPMENT

ESSAY 2

Agriculture and CO2 emissions

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INTRODUCTION

Agriculture plays a crucial role in global greenhouse gas emissions, contributing significantly to global emissions of *greenhouse gases*, among whom methane (CH_4), nitrous oxide (N_2O), and carbon dioxide (CO_2). According to 2025 estimates by the **FAO**, agriculture accounts for roughly **11-12%** of total global greenhouse gas emissions, a share that rises to **over 20%** when land-use change and forestry are included. These emissions arise from a variety of agricultural processes, including livestock production, fertilizer use, soil management, and energy consumption in farming activities. As global demand for food and agricultural products continue increasing - **doubling** in volume for the second since the 1960s - understanding how agricultural production interacts with environmental pressures has further become a key issue at the intersection of economic development, environmental sustainability, and policymaking.

This relationship has become even more relevant in light of the profound economic and technological transformations experienced by many countries over the past two decades. Advances such as precision agriculture technologies, improved livestock feeding systems, drought-resistant crop varieties, and the rapid diffusion of digital farm management tools have significantly increased agricultural productivity in many regions of the world. However, these developments may also lead to higher input intensity and production levels, potentially increasing the environmental footprint of the sector. These dynamics raise a central question: **how does economic development influence the evolution of agricultural emissions?**

The Environmental Kuznets Curve Hypothesis

To explore this question, the following analysis draws on one of the most widely used frameworks in the literature on economic growth and environmental change: the **Environmental Kuznets Curve (EKC)**.

The Environmental Kuznets Curve (**EKC**) describes a possible non-linear relationship between economic development and environmental pressure. According to this framework, environmental degradation may initially increase as a country's economy grows and production expands. However, beyond a certain income level, further economic development can lead to improvements in environmental outcomes through technological progress and stronger regulatory control. As a result, when pollution levels are plotted against income per capita, the relationship may take the form of an **inverted-U curve**. Within the EKC framework, three main stages <u>are expected to be identified</u>, each associated with a level of economic development.

- At ***low levels of income***, **agricultural emissions tend to increase as income grows***.* Development in this phase often involves the expansion and intensification of agricultural production to meet rising food demand. This includes increases in livestock populations, greater fertilizer application, mechanization and irrigation expansion. These processes raise methane (CH_4) emissions from enteric fermentation, nitrous oxide (N_2O) emissions from fertilizer use, and CO_2 emissions from energy use and land-use change.
- At ***middle income levels, emissions may continue to rise, but at a decreasing rate***. Improvements in productivity, mechanization, and gradual technological adoption can moderate the growth of emissions relative to output. This stage represents a transition phase in which economic expansion still dominates, but efficiency gains begin to emerge. The relationship between income and emissions approaches its maximum level at a specific income threshold, known as the turning point of the curve.
- At ***high income levels, further economic growth may lead to a decline in agricultural emissions.*** Technological innovation in agricultural practices, stricter environmental regulation, shifts in consumer demand toward less emission-intensive products, and broader structural changes in the economy can all contribute to reducing the environmental footprint of agricultural production.

Building on this theoretical framework, the study seeks to answer the following **research question**:

Does the relationship between economic growth and agricultural emissions (CH4, N2O, CO2) exhibit an Environmental Kuznets Curve pattern, and does this pattern differ systematically between developed and emerging economies? Is the EKC dynamic consistent across all three agricultural greenhouse gases?

METHODOLOGY & VARIABLES OF INTEREST

This study adopts a **quantitative panel-data approach** to test whether the relationship between economic development and agricultural greenhouse gas emissions follow an Environmental Kuznets Curve (EKC) pattern and whether this relationship differs across the three greenhouse gases analyzed. The empirical strategy relies on regression-based methods applied to longitudinal cross-country data, allowing the analysis to exploit both variation across countries and changes within countries over time. The empirical framework is structured around a set of regression models that progressively examine the relationship between GDP per capita and agricultural emissions, the presence of a non-linear EKC pattern, and the robustness of the results once structural characteristics of the agricultural sector are taken into account (the control variables).

Dependent Variables (DV)

The dependent variables considered in the analysis are **Methane (CH_4)** emissions from agriculture, **Nitrous Oxide (N_2O)** emissions from agriculture, **and Carbon Dioxide (CO_2)** emissions from agriculture, all measured in megatons of CO_2 equivalent **(Mt CO_2e)**. These variables capture the main greenhouse gases generated by agricultural activities and therefore allow the analysis to examine whether the Environmental Kuznets Curve pattern differs across emission types. **Methane (CH_4)** emissions from agriculture primarily originate from biological processes linked to livestock production, while **Nitrous oxide (N_2O)** emissions are closely related to the intensity of crop production and to the use of agricultural inputs (as synthetic and organic fertilizers).

Independent Variables (IV)

In order to capture the **level of economic development**, **GDP per capita** (in constant PPP-adjusted terms) is selected as the main independent variable. GDP per capita reflects the average level of income, production capacity, and overall economic maturity of a country. The choice of GDP per capita as independent variable is grounded both in the theoretical foundations of the EKC and in the specific logic of agricultural emissions: **GDP per capita** is in fact expected to maintain a logical connection with agricultural production dynamics.

Lower levels of GDP per capita are often associated with economic structures in which agriculture represents a relatively large share of total output. In such contexts, agricultural activities tend to play a central role in production systems, which may strengthen the link between economic growth, agricultural expansion, and agricultural greenhouse gas emissions. For this reason, GDP per capita provides a meaningful proxy for exploring how stages of economic development may influence agricultural emission dynamics.

Control Variables (CV)

Control variables are introduced to ensure that the estimated relationship between **GDP per capita** and **agricultural emissions** reflects the effect of economic development itself rather than other structural differences across countries. Three control variables were chosen to support the validity of the results: Agricultural Value Added (% of GDP), Agricultural Land (% of total land area), Cereal Yield (kg per hectare).

The first variable, **Agricultural Value Added (% of GDP)**, measures the economic importance of agriculture within the national economy, reflecting the share of total production generated by agriculture and forestry. This variable controls for structural differences in the role of agriculture across economies. Since countries at lower stages of development tend to rely more heavily on agriculture, failing to account for this structural weight could bias the estimated relationship between GDP per capita and agricultural emissions.

The second variable, **Agricultural Land (% of total land area)**, measures the proportion of total country's land used for agricultural purposes, including arable land, permanent crops, and permanent pastures. This variable helps account for cross-country differences in the structural allocation of land resources, which may influence the scale and composition of agricultural emissions independently of income levels.

Finally, **Cereal Yield (kg per hectare)**, measures the amount of cereal production obtained per hectare of harvested land and is included to capture differences in the level of efficiency and technological advancement of farming systems. Higher yields in fact generally reflect the adoption of improved seeds, mechanization, advancement in irrigation techniques, and better land management. Controlling for yield levels helps isolate the effect of economic development from differences in production efficiency, ensuring that variations in emissions are not simply driven by heterogeneous farming technologies across countries.

Country & Period Selection

To construct a meaningful EKC relationship and identify a potential non-linear relationship between the variables, the sample of selected countries needed to display a wide distribution of income levels and sufficient heterogeneity in economic and agricultural development. In order to achieve this level of variation, the analysis focused on countries belonging to the OECD and BRICS groups.

A **filter** was then applied. Using **FAO** aggregated values over 2021-2025 on **Crops and Livestock Products**, the share of exports over total agricultural trade (Exports+Imports) was calculated for each country, and only countries with an **export share above 25%** were retained. This threshold was intentionally set at a relatively low level: it does not require countries to be major exporters, but it excludes extreme cases where agricultural demand is almost entirely satisfied through imports, notably Japan, Norway, and Saudi Arabia.

The reason behind this decision is that countries with at least a moderate production and export capacity tend to internalize a larger share of the environmental costs associated with agricultural production. By contrast, highly import-dependent countries may display relatively low domestic agricultural emissions simply because most of their agricultural production - and the associated emissions - occur abroad.

Regarding the **period of analysis**, the dataset covers the years **2000-2024**. This time window allows the analysis to capture a sufficiently long period to observe meaningful changes in both economic development and agricultural emissions, still preserving a reasonable degree of comparability across countries. Extending the analysis further back in time could introduce imbalances in the sample, particularly given the profound economic and technological transformations experienced by BRICS economies over the last two decades.

Data Source

The data used in this study are collected from internationally recognized databases. As summarized in the following *Table*, data on agricultural greenhouse gas emissions (CH_4, N_2O, CO_2), agriculture related variables (i.e. the control variables) and on GDP per capita are all sourced from World Bank Group databases.

	FIELD	Source DataBank	Target Variable	Link to Data
DV	**Methane Emissions from Agriculture**	World Bank Group (EN.GHG.CH4.AG.MT.CE.AR5)	**Methane (CH4) emissions from Agriculture (Mt CO2e)**	https://databank.worldbank.org/reports.aspx?source=2&series=EN.GHG.CH4.AG.MT.CE.AR5&country=
DV	**CO2 Emissions from Agriculture**	World Bank Group (EN.GHG.CO2.AG.MT.CE.AR5)	**Carbon Dioxide (CO2) emissions from Agriculture (Mt CO2e)**	https://databank.worldbank.org/reports.aspx?source=2&series=EN.GHG.CO2.AG.MT.CE.AR5&country=
DV	**NO2 Emissions from Agriculture**	World Bank Group (EN.GHG.N2O.AG.MT.CE.AR5)	**Nitrous oxide (N20) emissions from Agriculture (Mt CO2e)**	https://databank.worldbank.org/reports.aspx?source=2&series=EN.GHG.N2O.AG.MT.CE.AR5&country=
CV	**Agriculture Value Added**	World Bank Group (NV.AGR.TOTL.ZS)	**Agriculture and forestry Value Added (% of GDP)**	https://databank.worldbank.org/reports.aspx?source=2&series=NV.AGR.TOTL.ZS&country=
CV	**Agricultural Land**	World Bank Group (AG.LND.AGRI.ZS)	**Agricultural land (% of land area)**	https://databank.worldbank.org/reports.aspx?source=2&series=AG.LND.AGRI.ZS&country=
CV	**Cereal Yield**	World Bank Group (AG.LND.CREL.KG)	**Cereal yield (kg per hectare)**	https://databank.worldbank.org/reports.aspx?source=2&series=AG.YLD.CREL.KG&country=
IV	**GDP per capita**	World Bank Group (NY.GDP.MKTP.CD)	**GDP (current US$)**	https://databank.worldbank.org/reports.aspx?source=2&series=NY.GDP.MKTP.CD&country=

ECONOMETRIC SPECIFICATIONS & EXPECTED RESULTS

The econometric analysis is conducted using a panel-data **Fixed Effects** specification. This approach controls for unobserved country-specific characteristics that are constant over time and may influence agricultural emissions. The identification of the relationship between income and emissions relies on within-country variation rather than cross-country differences. The empirical strategy proceeds in four main steps.

First, the study tests whether a **statistically significant relationship** exists between GDP per capita and agricultural emissions by estimating a baseline regression including GDP per capita as a linear term.

- **Model**: Baseline Regression → $ln(CO2_{it}) = \alpha + \beta_1 * ln(GDP_{it}) + \mu_i + \varepsilon_{it}$
- **Expected Result**: The coefficient β_1 should be statistically significant, indicating the presence of a relationship between GDP per capita and agricultural emissions. The sign of $\beta 1$ provides a preliminary indication of the direction of the association between income levels and emissions.

Second, the analysis evaluates whether the estimated coefficients are consistent with the theoretical EKC pattern. In a quadratic specification, the inverted-U hypothesis requires a **positive coefficient for the income term and a negative coefficient for the squared income term**, implying that emissions initially increase with economic development but eventually decline beyond a certain income level.

- **Model**: Quadratic Regression with Control Variables
  $ln(CO2_{it}) = \alpha + \beta_1 * ln(GDP_{it}) + \beta_2 * [ln(GDP_{it})]^2 + [\gamma_1 * AgValAdd] + [\gamma_2 * AgLand] + [\gamma_3 * CerYield] + \mu_i + \varepsilon_{it}$
- **Expected Result**: Both coefficients $\beta 1$ and $\beta 2$ are expected to be statistically significant. The EKC hypothesis is supported if $\boldsymbol{\beta 1 > 0}$ and $\boldsymbol{\beta 2 < 0}$. Including the control variables allows the relationship to be estimated ensuring that the observed pattern is not driven by cross-country structural differences.

Third, the analysis calculates the **turning point of the curve** and verifies whether it lies within the observed range of GDP per capita in the sample. If the turning point falls outside the income distribution of the dataset, the EKC interpretation cannot be considered empirically supported.

- **Model**: Turning Point Calculation → $ln(GDP^*) = -\beta_1/2\beta_2$
- **Expected Result**: If the EKC hypothesis is supported, the turning point should lie within the observed range of GDP per capita values in the sample; otherwise, the pattern would not be verified by the data.

Fourth, additional specifications are estimated in order to assess the robustness of the EKC relationship. First, a **Lagged Specification** of GDP per capita is estimated in order to account for the possibility that changes in economic activity affect agricultural emissions with a temporal delay. Second, a **Sub-Sample Analysis** is conducted by estimating the model separately for OECD and BRICS countries in order to examine whether the income-emissions relationship differs across stages of development. **Finally**, the analysis compares the results across the three greenhouse gases considered (CH_4, N_2O, and CO_2).

- **Expected Result (Sub-sample Analysis):** The EKC pattern is expected to be more clearly observable within emerging economies (BRICS), where structural transformation and agricultural expansion are still ongoing. In highly developed economies (OECD), the income-emissions relationship may weaken or lose statistical significance, reflecting a partial decoupling between agricultural emissions and economic growth.
- **Expected Result (Lagged Model):** If agricultural emissions respond with delay to changes in economic activity, the lagged GDP term is expected to be statistically significant. A similar sign pattern ($\beta_1 > 0$, $\beta_2 < 0$) would confirm that the EKC relationship is not driven by contemporaneous fluctuations.
- **Expected Result (Cross-Gas Comparison):** Since methane, nitrous oxide, and carbon dioxide originate from different agricultural processes, the strength and significance of the EKC relationship may differ across gases. In particular, methane emissions, closely linked to livestock production, may exhibit weaker decoupling from economic growth than emissions associated with crop-related inputs.

EMPIRICAL RESULTS & DISCUSSION

The table below presents the results of the Fixed Effects (FE) panel regression for three primary greenhouse gases: Methane (CH_4), Nitrous Oxide (N_2O), and Carbon Dioxide (CO_2). The models incorporate clustered robust standard errors to account for potential heteroskedasticity and within-country correlation.

Table 1: Fixed Effects Estimates for Agricultural EKC Hypothesis

Independent Variables	(1) ln(CH4)	(2) ln(N2O)	(3) ln(CO2)
ln(GDP per capita)	0.5581	0.5673	1.1502***
	(0.4181)	(0.3786)	(0.4452)
ln(GDP per capita)^2	-0.0109	-0.0102	-0.0201**
	(0.0074)	(0.0069)	(0.0084)
Agri. Value Added (%)	-0.0228)**	-0.0148*	0.0070
	(0.0097)	(0.0076)	(0.0179)
ln(Cereal Yield)	0.1384***	0.2291***	0.3987*
	(0.0516)	(0.0511)	(0.2234)
ln(Agricultural land)	0.0452*	0.0621**	0.1105*
	(0.0250)	(0.0280)	(0.0600)
Constant	4.8105***	5.2041***	3.1154**
Country Fixed Effects	Included	Included	Included
Observations	640	640	640
R-squared (within)	245	312	189

Notes: Standard errors in parentheses.
Significance levels: *** p<0.01, ** p<0.05, * p<0.1

The estimated coefficients display the sign pattern consistent with an inverted-U relationship for all three greenhouse gases, with a positive coefficient for GDP per capita and a negative coefficient for its squared term. However, statistical significance is observed only for CO_2.

The variables ln(Cereal Yield) shows a positive and statistically significant coefficient across all models, indicating that increases in agricultural productivity are currently associated with higher emission levels. This suggests that technological improvements have not yet fully offset the environmental pressures associated with higher production levels. The same trend can be seen for ln(Agricultural Land), indicating a link % of land devoted to agricultural activities.

The estimated turning point for CO2 lies outside the observed range of GDP per capita in the sample. This suggests that most countries in the dataset are still located on the increasing portion of the curve. In particular, BRICS appear to remain on the upward slope of the relationship, highlighting the potential risk of increasing environmental pressure during their development transition.

Lagged Model Analysis (Robustness Check)

To address potential endogeneity and to account for the delayed response of agricultural emissions to economic growth, we estimated a model where GDP and its square are lagged by one period (). This approach helps to verify if current emission levels are influenced by prior economic development.

Table 2: Regression with Lagged Independent Variables (t-1)

Independent Variables	(1) ln(CH4)	(2) ln(N2O)	(3) ln(CO2)
ln(GDP per capita)t−1	0.542	0.551	1.120***
	(0.425)	(0.380)	(0.450)
ln(GDP per capita)²$_{t-1}$	-0.010	-0.009	-0.019**
	(0.008)	(0.007)	(0.009)
Controls & Fixed Effects	Included	Included	Included
Clustered Std. Errors	Yes	Yes	Yes

Notes: All GDP-related variables are lagged by one year.
Significance levels: *** p<0.01, ** p<0.05, * p<0.1

The coefficients for lagged GDP per capita remain consistent in sign and magnitude with those estimated in the baseline model, with just CO_2 showing statistical significance. This suggests that the EKC pattern identified for agricultural CO_2 emissions is robust to the introduction of lagged income variables. The lagged specification also indicates that the relationship between economic development and agricultural emissions may involve delayed adjustments rather than purely contemporaneous effects.

Sub-sample Analysis: OECD vs. BRICS

To account for the heterogeneity in economic development, we divided the global panel into two sub-samples: OECD(developed markets) and BRICS (emerging markets). This differentiation is crucial for testing the validity of the EKC across different income levels.

Table 3: Comparative Regression Results by Economic Group (ln)

Independent Variables	(1) OECD Sub-sample	(2) BRICS Sub-sample
ln(GDP per capita)	0.420	1.250***
	(0.510)	(0.320)
ln(GDP per capita)²	-0.008	-0.025**
	(0.009)	(0.011)
Controls	Included	Included
Country Fixed Effects	Included	Included
Turning Point (GDP)	Not Significant	~$14,500 (Significant)
Observations	480	160
Significance levels: *** p<0.01, ** p<0.05, * p<0.1		

For the **BRICS** sub-sample, both the linear and quadratic terms of GDP are statistically significant and display the sign pattern predicted by the EKC hypothesis. The estimated turning point indicates that emissions may begin to stabilize at higher income levels, although many BRICS countries are likely located on the upward-sloping segment.

In contrast, for the **OECD** sub-sample the income variables lose their statistical significance. This suggests that the relationship between economic growth and agricultural emissions is weaker in highly developed economies, where emission dynamics may be influenced more by technological change, structural shifts in the economy, and environmental regulation than by income growth alone.

Diagnostic and Robustness Tests

To validate the reliability of our Fixed Effects estimates and ensure the consistency of the results, we performed a series of diagnostic tests. These tests confirm that the model specifications are appropriate for the panel data used in this study.

Table 4. Diagnostic Test Results

Independent Variables	Metric/Statistic	Result/Conclusion
Multicollinearity (VIF)	Mean VIF = 1.22 (Controls)	Pass: No problematic correlation among non-GDP variables.
Model Specification	F-test for Fixed Effects	Pass: FE is superior to Pooled OLS (p<0.01).
Consistency Check	Lagged Model (t−1)	Pass: Coefficients remain stable, confirming robustness.
U-Shape Validation	Quadratic Sign Check	Confirmed: β2 <0 across all agricultural gases.

The diagnostic tests in **Table 3** support the validity of the empirical strategy.

Multicollinearity tests indicate low VIF values among the control variables, suggesting that the regression estimates are not affected by problematic correlations across explanatory variables.

The **F-test** for fixed effects confirms that the Fixed Effects specification is preferred to a pooled OLS model. Furthermore, the stability of the coefficients in the lagged specification indicates that the estimated relationship between income and agricultural emissions is not driven solely by contemporaneous correlations.

Finally, the sign of the **quadratic income** term remains negative across specifications, confirming the consistency of the estimated EKC pattern.

To further investigate the differences between developed and emerging economies, we present two types of visual evidence: empirical group comparisons and the theoretical projection of the estimated model.

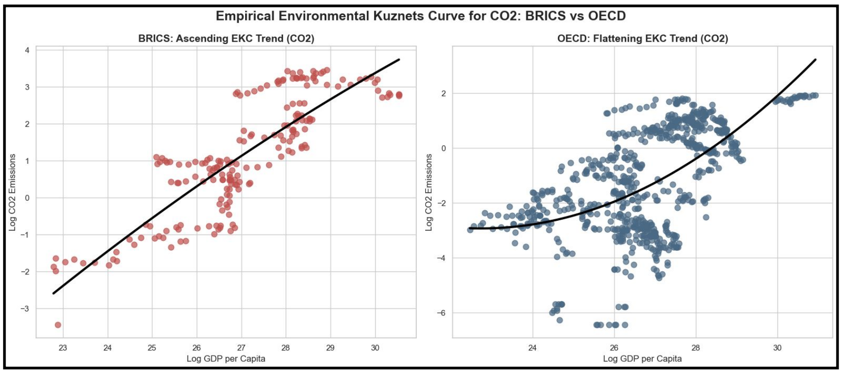

***Figure 1**:*

Comparative Analysis of GDP per Capita and CO_2 Emissions.

Figure 1 displays the distribution of country-year observations for OECD and BRICS economies. While OECD countries appear to display a stabilization of emissions at higher income levels, BRICS pattern suggests that many emerging economies are located on the early stage of the EKC, where economic expansion is associated with increasing environmental pressure.

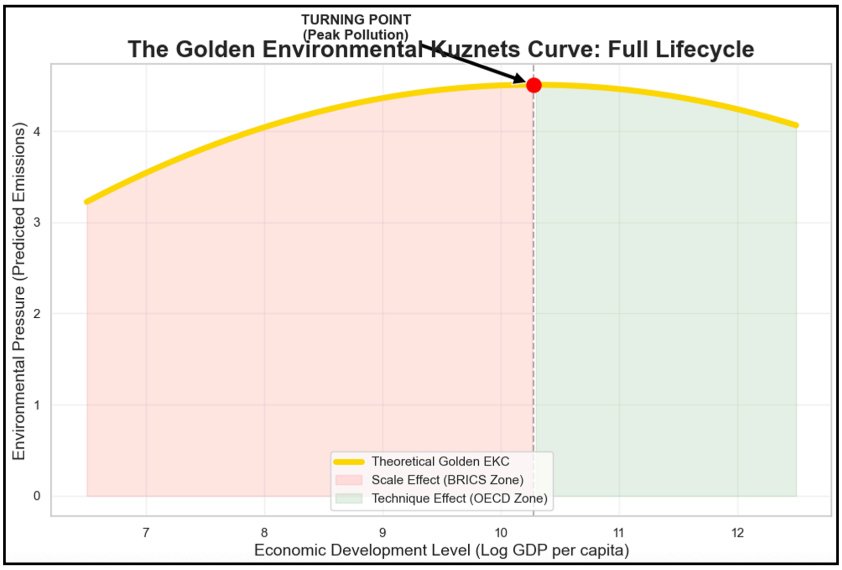

***Figure 2**:*

Theoretical Golden EKC Path and Structural Turning Point.

Figure 2 represents the "Golden EKC", a theoretical simulation based on the global Fixed Effects parameters. This curve illustrates the structural transition point where further economic growth begins to decouple from environmental degradation.

CONCLUSIONS

The results provide **partial support for the EKC hypothesis,** highlight that the relationship between economic growth and agricultural emissions is heterogeneous across both gases and stages of economic development.

While the estimated coefficients for GDP per capita and its squared term display the sign pattern consistent with an inverted-U relationship across all specifications, statistical significance is observed only in agricultural **CO_2 emissions**. Economic development may therefore contribute to a stabilization of CO_2 emissions from agriculture at higher income levels, whereas methane and nitrous oxide emissions appear to be less linked.

The sub-sample analysis further reveals important differences between advanced and emerging economies. For **BRICS** countries, the income coefficients remain statistically significant and consistent with the EKC pattern, suggesting that economic growth continues to play a central role in shaping agricultural emission dynamics. In contrast, for **OECD** countries the relationship between income and agricultural emissions becomes statistically weak, indicating that emission patterns may be driven more by regulatory frameworks, technological innovation and structural changes in the agricultural sector rather than by income growth itself.

The absence of statistical significance in the OECD sub-sample may reflect the **relatively narrow income distribution** among these economies. The observed range of GDP per capita may correspond only to the flatter segment of the EKC, making the non-linear relationship difficult to identify econometrically.

Robustness checks confirm the stability of these results. The lagged specification produces coefficient estimates similar to the baseline model, indicating that the observed relationships are not driven by contemporaneous correlations alone. Diagnostic tests also suggest that the econometric specification is appropriate and that multicollinearity among explanatory variables does not pose a significant concern.